Amirkabir University of Technology
(Tehran Polytechnic)

Department of Electrical Engineering

Report for the Electronics 1 project
Phase 1

5 V Regulated Power Supply

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In the first phase of the project, we are required to design a circuit that, with an input voltage of 8 V amplitude and 500 Hz frequency, produces a constant 5 V output with a maximum ripple of 0.05 V.

We start with a simple AC to DC converter circuit. To better display the output in this case, we use ADS software for simulation.

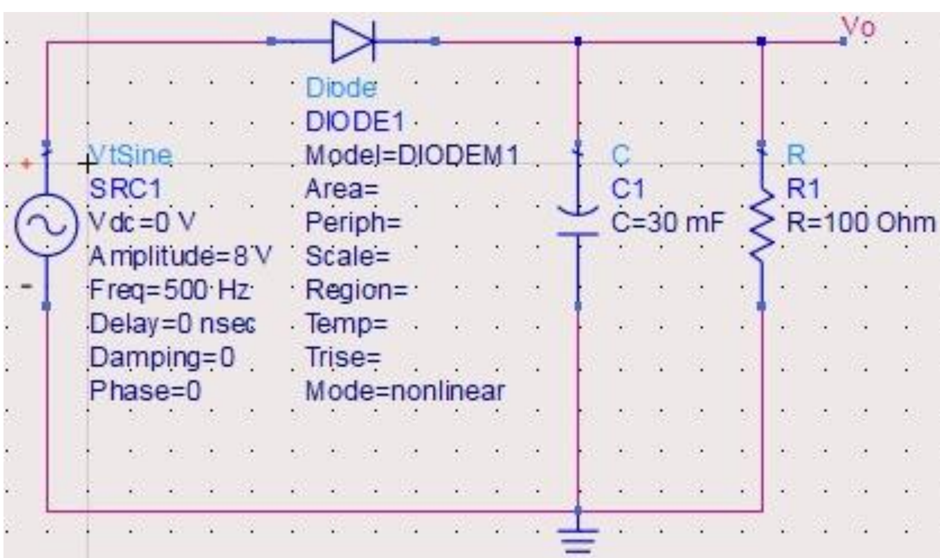
Given that the output resistance is 100 ohms as stated in the problem, only the capacitor value remains to be calculated. Therefore, using the approximate ripple formula:

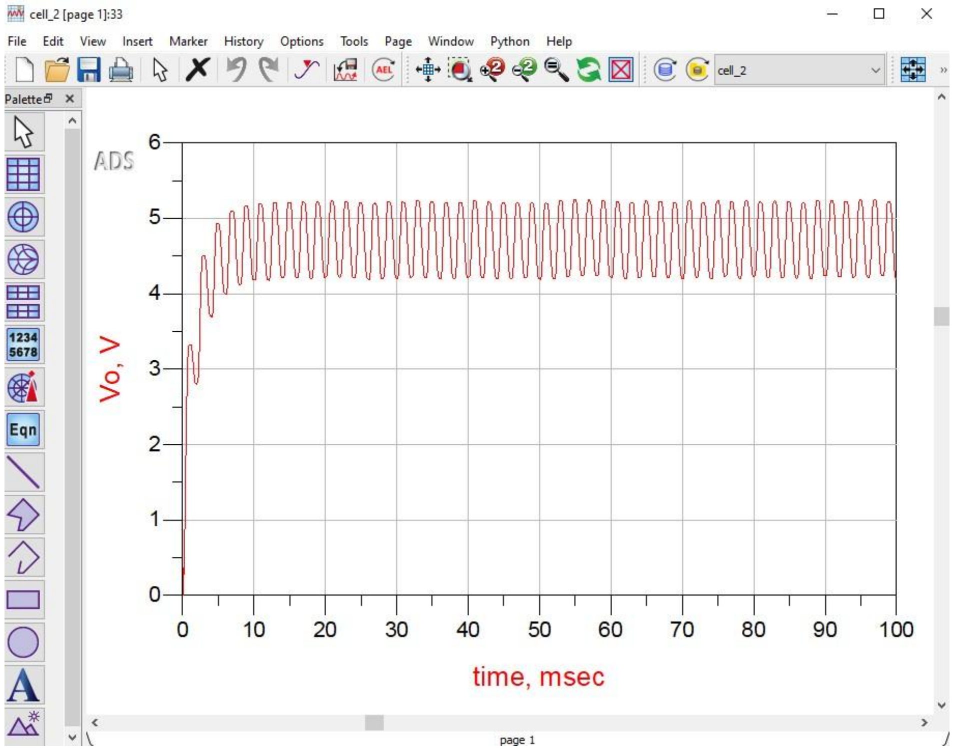
$$V_R \approx \frac{V_p - V_{D,on}}{0.5 + R_L Cf}$$

$$0.05 > \frac{8 - 0.5}{0.5 + (100)(\mathrm{C})(500)}$$

$$C > 30\,mF$$

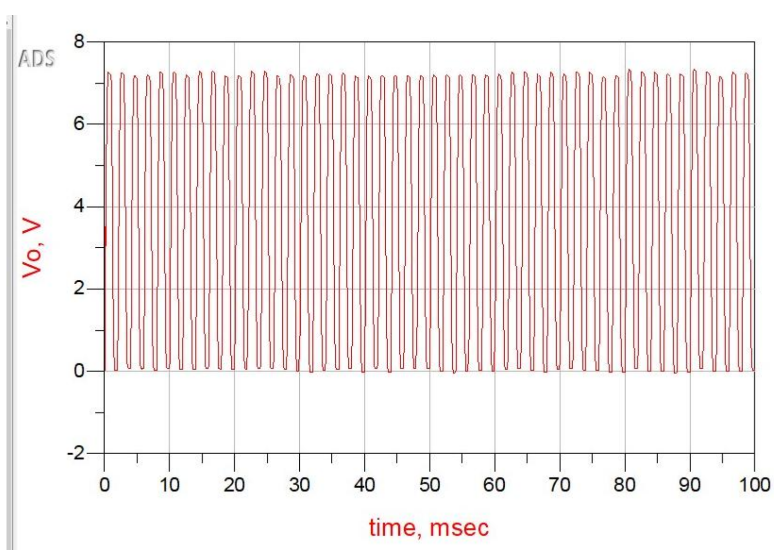
The calculated capacitor value is very large, and by inserting this value into the circuit, we get:





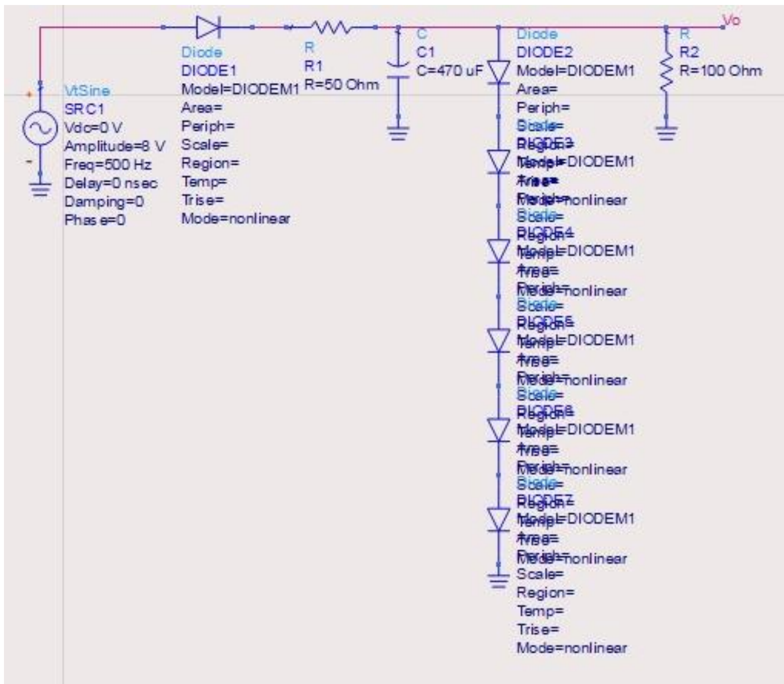
By increasing the capacitor value (for example, 470 mF), the ripple amount also decreases. However, since we have a limitation on the capacitor value we can use, we employ the maximum value that can be utilized in this project, which is 470 µF.

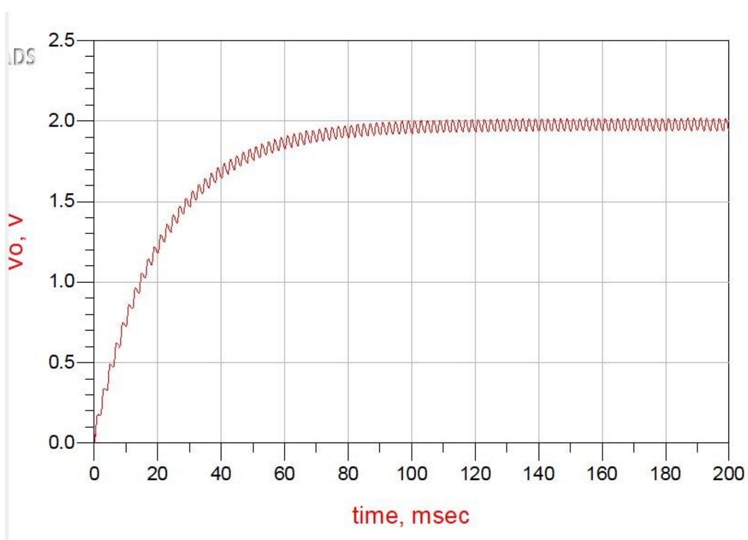
(The maximum usable capacitance in this project is 500 µF, but since in reality and practically a 470 µF capacitor is available, we use this value.)
In this case, the output becomes as follows:



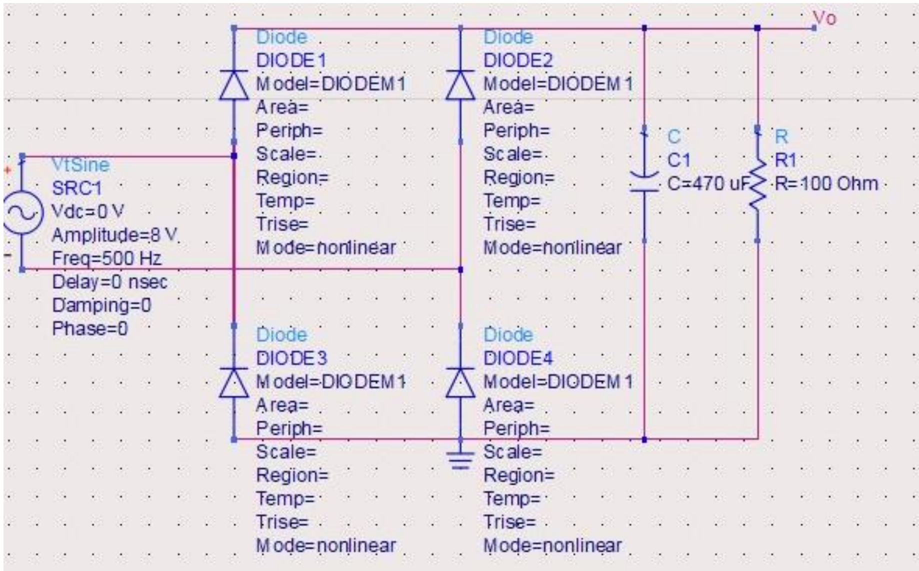
So we must use another method to reduce the ripple.

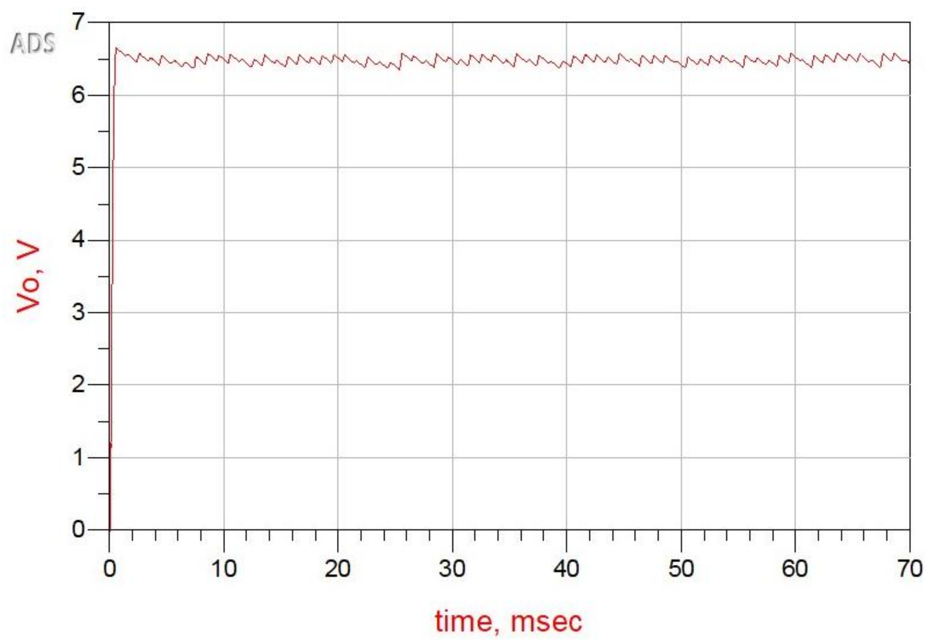
We can use either a voltage regulator or a diode bridge. Here we use a diode bridge. The reason is that in a voltage regulator, the output voltage drops by the sum of the bias voltages of the diodes used relative to the input voltage. This voltage drop is less than the 5V output and causes problems:



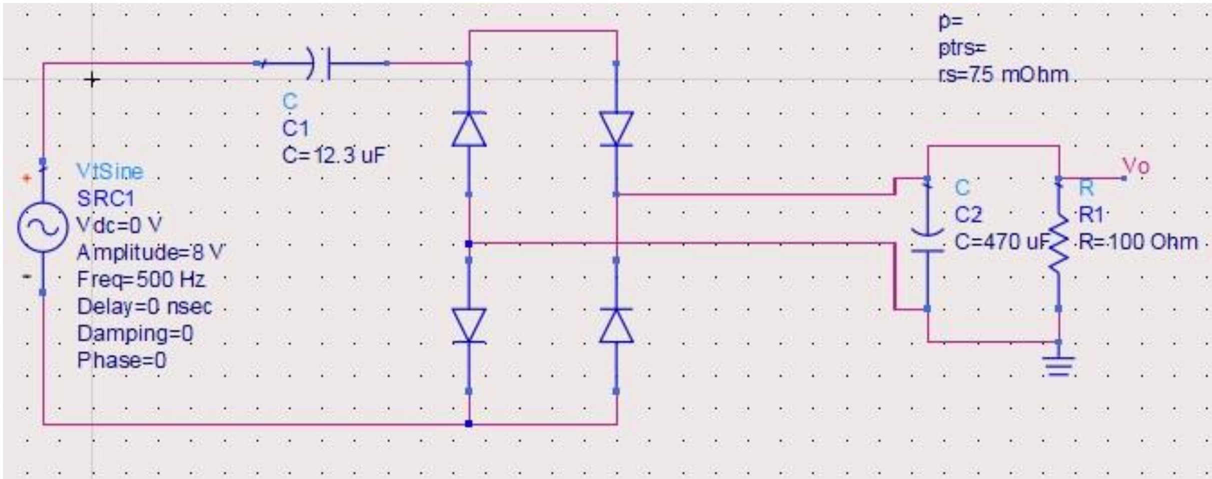


To increase this constant output voltage, it is enough to use more diodes, which is relatively difficult. Therefore, we prefer the diode bridge:

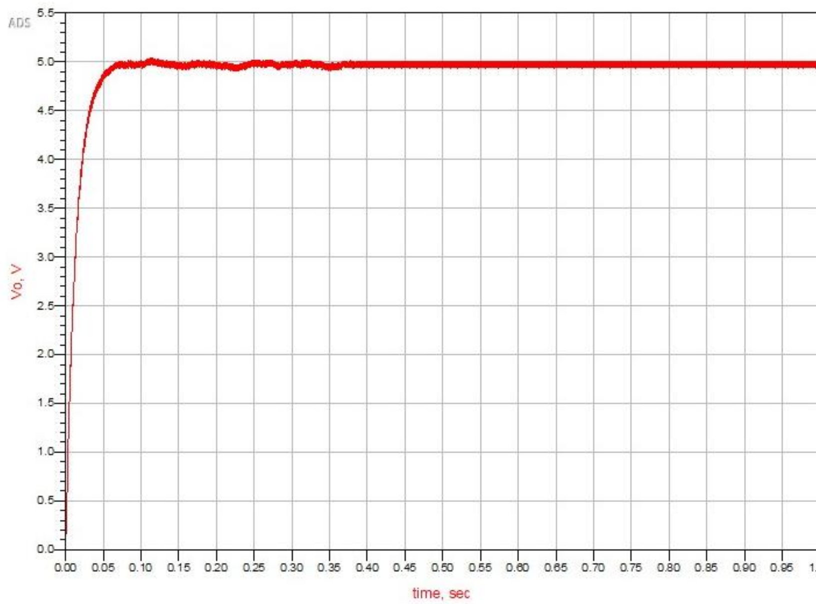




Now it is sufficient to use a capacitor at the input so that the constant output voltage becomes smaller and the ripple voltage becomes more periodic. By changing the input capacitor value, we arrive at this circuit:



The output is as follows:



In a closer view, we have: